

AFFINE TORSION COVERS OF THE $(3, 4, \infty)$ PERIOD SYSTEM AND EXACT ERDŐS–STRAUS DENOMINATOR DEFECTS

THE CLANKERS: SOURCE-ADDRESSABLE MATHEMATICAL NOTE

ABSTRACT. For the specified integral period monodromy we compute a three-coordinate affine slice at every odd torsion level. Its extension cocycle has exact order equal to that level. We prove a complete orbit classification, with one orbit away from three and two orbits at levels divisible by three, and give closed genus formulas. The return denominator of each fully marked Erdős–Straus divisor candidate maps faithfully to this cocycle subgroup. The comparison uses the actual source matrices, not the source’s global six-sphere claim. No universal ES positivity theorem or historical priority is asserted.

1. THE SOURCE REPRESENTATION AND ITS AFFINE SLICE

We use the following integral matrices on the ordered rank-four lattice $\mathbb{L} = \mathbb{Z}\hat{\gamma} \oplus \mathbb{Z}\hat{u} \oplus \mathbb{Z}\hat{w} \oplus \mathbb{Z}\hat{\delta}$:

$$A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 6 & 0 & 1 & 0 \\ -6 & -1 & -1 & 0 \\ -2 & 1 & 0 & 1 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -6 & 1 & 0 & 0 \\ 3 & 0 & 1 & 1 \end{pmatrix}, \quad A_\infty = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}. \quad (1)$$

They are the inverse-transpose monodromies in the circulated $(3, 4, \infty)$ period construction. The source locator is [Source(2026), §2], Lemmas 2.4–2.7. Direct multiplication proves $A_1^3 = A_2^4 = A_1 A_2 A_\infty = I$. We distinguish $\mathbb{L}$ from the rank-24 Leech lattice. No global assertion that the source threefold is S^6 is used below.

The invariant covector γ reads the first coordinate. For a positive odd integer D , let

$$\mathcal{F}_D = \{(1, x, y, z) \in (\mathbb{Z}/D)^4\} \cong (\mathbb{Z}/D)^3.$$

The induced permutations, composed right to left, are

$$\begin{aligned} a_1(x, y, z) &= (6 + y, -6 - x - y, z - 2 + x), \\ a_2(x, y, z) &= (-y, x - 6, z + 3 + y), \\ a_\infty(x, y, z) &= (x, y + x, z - 1), \end{aligned} \quad (\text{mod } D). \quad (2)$$

Their inverses are a_1^2, a_2^3 , and $(x, y, z) \mapsto (x, y - x, z + 1)$. The unique residue class modulo one is allowed; $\mathcal{F}_1$ is a singleton.

The base marking used earlier in the Fable comparison is reached by the explicit projective change $s = 1/2 - t$, with inverse $t = 1/2 - s$. It sends $(0, 1, \infty)$ to $(1/2, -1/2, \infty)$ and transports the three loop actions with those labels. It changes neither the modulus nor the monodromy representation. It is not a substitution into a Mellin integral.

Theorem 1.1 (Primitive obstruction class). *Write $A_g = \begin{pmatrix} 1 & 0 \\ b_g & B_g \end{pmatrix}$ for $g \in \Delta = \langle g_1, g_2 \mid g_1^3 = g_2^4 = 1 \rangle$. Then $b_{gh} = b_g + B_g b_h$. The integral cocycle class $[b]$ has infinite order and spans a split $\mathbb{Z}$ -summand of $H^1(\Delta, \mathbb{Z}_B^3)$. Its reduction $[b_D] \in H^1(\Delta, (\mathbb{Z}/D)_B^3)$ has exact additive order D . It restricts to zero on each finite cyclic subgroup $\langle g_1 \rangle$ and $\langle g_2 \rangle$.*

The cover of $B^\circ = \mathbb{P}^1(\mathbb{C}) \setminus \{0, 1, \infty\}$ with fibre $\mathcal{F}_D$ and these loop actions has a section exactly when $D = 1$.

Proof. The cocycle identity is block multiplication. If a cocycle is changed by a coboundary $B_g v - v$, its third coordinate at $g_\infty = (g_1 g_2)^{-1}$ does not change: the third row of $B_\infty - I$ is zero. Thus evaluation of that coordinate gives a homomorphism from H^1 to $\mathbb{Z}$, or to $\mathbb{Z}/D$. Its value on $[b]$ is -1 . Over $\mathbb{Z}$ the map $n \mapsto n[b]$ is therefore a split injection. Modulo D a multiple $k[b_D]$ can vanish only if $D \mid k$, while $D[b_D] = 0$ coefficientwise. The local fixed vectors $(2, -4, 0)$ and $(3, -3, 0)$ solve $b_j + B_j v = v$ for $j = 1, 2$, proving the two local trivializations.

A section of a finite covering over the connected base is a common fixed sheet of all monodromies: a chosen value continues uniquely along paths, and is single-valued precisely under that condition. The cusp

changes z by -1 , so no fixed sheet exists for $D > 1$; for $D = 1$ the sole sheet is fixed. Equivalently an affine fixed point trivializes the cocycle. $\square$

This exhibits a locally split but globally nontrivial three-coordinate system, without replacing its arithmetic modulus by a continuous parameter. The source finite-filling maps $x \mapsto A_j x + v_j/m_j$ on a real torus are *different maps*: (2) is the linear monodromy restricted to $\gamma = 1$ modulo D , not a substitution for those filling translations.

2. ALL ORBITS AND THEIR COMPACTIFIED GENERA

Lemma 2.1 (Two shears and their central commutator). *The source words $H = [A_2^2, A_1]$ and $V = A_2 H A_2^{-1}$ induce*

$$H(x, y, z) = (x, y + 6, z + x - 6), \quad V(x, y, z) = (x - 6, y, z + y + 6).$$

Their commutator is

$$H V H^{-1} V^{-1}(x, y, z) = (x, y, z - 12). \quad (3)$$

These are identities over $\mathbb{Z}$, and hence in every residue ring.

Proof. Multiplication in (1) gives

$$H = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 6 & 0 & 1 & 0 \\ -6 & 1 & 0 & 1 \end{pmatrix}, \quad V = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -6 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 6 & 0 & 1 & 1 \end{pmatrix}.$$

Their inverse affine formulas are $(x, y - 6, z - x + 6)$ and $(x + 6, y, z - y - 6)$. Substitution proves (3). $\square$

Theorem 2.2 (Complete odd-level classification). *If $3 \nmid D$, the action on $\mathcal{F}_D$ is transitive. If $3 \mid D$, it has exactly two orbits*

$$O_0 = \{x = y = 0 \pmod{3}\}, \quad O_1 = \mathcal{F}_D \setminus O_0,$$

of sizes $D^3/9$ and $8D^3/9$. The associated compact connected branched covers of $\mathbb{P}^1$ have genera

$$g(D) = \frac{5D^3 - 12D^2 - 17D + 24}{24}, \quad (3 \nmid D), \quad (4)$$

$$g_0(D) = \frac{5D^3 - 12D^2 - 81D + 216}{216}, \quad g_1(D) = \frac{5D^3 - 12D^2 - 9D + 27}{27} \quad (3 \mid D). \quad (5)$$

Every cusp cycle has length exactly D .

Proof. If $3 \nmid D$, both 6 and 12 are units modulo the odd integer D . The powers of H, V reach arbitrary x, y , and their central commutator then adjusts z arbitrarily. This proves transitivity with no assumption that D is prime.

Suppose $3 \mid D$. Modulo three the planar actions are $(x, y) \mapsto (y, -x - y)$ and $(x, y) \mapsto (-y, x)$. They fix zero and are transitive on the eight nonzero pairs. For an explicit verification, the words

$$1, a_1, a_2, a_2 a_1, a_2^2, a_2^2 a_1, a_2^3, a_2^3 a_1$$

send $(0, 1)$ respectively to $(0, 1), (1, 2), (2, 0), (1, 1), (0, 2), (2, 1), (1, 0), (2, 2)$. The powers of H, V then realize every planar change divisible by three, since $6(\mathbb{Z}/D) = 3(\mathbb{Z}/D)$. It remains to adjust z while restoring x, y . When $x = 0 \pmod{3}$, choose n with $6n = -x \pmod{D}$. The word $H^n a_\infty$ fixes x, y and shifts z by $c = -1 + n(x - 6)$, which is $-1 \pmod{3}$. Together with the central shift -12 , this generates every z -translation, because $\gcd(c, 12, D) = 1$. For a nonzero planar residue first use the listed words to reach $(0, 1)$ modulo three, apply this adjustment, and conjugate back. The two indicated orbits are thus complete.

A cusp iterate is $(x, y + nx, z - n)$, so its length is D . Solving the fixed-point equations gives

$$\text{Fix}(a_1) = \{(2, -4, z)\}, \quad \text{Fix}(a_2) = \text{Fix}(a_2^2) = \{(3, -3, z)\}.$$

The last equality follows from $a_2^2(x, y) = (-x + 6, -y - 6)$ and invertibility of 2. Thus a_2 has no cycles of length two. If $3 \nmid D$, the three cycle counts are

$$c_1 = (D^3 + 2D)/3, \quad c_2 = (D^3 + 3D)/4, \quad c_\infty = D^2.$$

If $3 \mid D$, all a_1 fixed points lie in O_1 and all a_2 fixed points in O_0 . For an orbit of size n with f_1, f_2 fixed points,

$$c_1 = f_1 + (n - f_1)/3, \quad c_2 = f_2 + (n - f_2)/4, \quad c_\infty = n/D.$$

Use $(n, f_1, f_2) = (D^3/9, 0, D)$ and $(8D^3/9, D, 0)$. The covering exists by the permutation-triple/Riemann-existence correspondence [Sijlsing and Voight(2014), Introduction and §1]; concretely, copies of a slit sphere are glued according to the permutations, and a length- e cycle is filled by the local map $w \mapsto w^e$. This also

specifies the complex structure. The bicoloured cell structure has $c_1 + c_2$ vertices, n edges, and c_∞ faces, so $2 - 2g = c_1 + c_2 + c_\infty - n$. Substitution proves (4)–(5). $\square$

The finite permutations and their cyclic order are a constructive specification of the covers. The certificate does not purport to give rational equations for every map, or to prove that every component is defined over $\mathbb{Q}$ rather than $\overline{\mathbb{Q}}$. For $D = 3$ the two components have degrees $(3, 24)$ and genera $(0, 1)$; for $D = 5$ there is one component of degree 125 and genus 11. These numbers are not identifications with the Albert algebra or Leech lattice.

Proposition 2.3 (Functorial modulus maps). *For $D \mid E$, reduction $\mathcal{F}_E \rightarrow \mathcal{F}_D$ is equivariant and surjective, with fibres of size $(E/D)^3$. For coprime odd D, E ,*

$$\mathcal{F}_{DE} \xrightarrow{\sim} \mathcal{F}_D \times \mathcal{F}_E$$

is equivariant in all three coordinates and all three monodromies. An inverse coordinate is $a + D((b - a)D^{-1} \bmod E) \bmod DE$. The cocycles, local systems and completed covers have the corresponding maps. Products are taken over the same base and the same candidate label.

Proof. All maps are integral polynomials, so commute with reduction. The coordinate CRT formula proves both inverse compositions and the fibre counts. Extension across a puncture follows in local coordinates: a cycle of length E maps to one of length D by $w \mapsto w^{E/D}$, with its marked branch included. No CRT assertion is made between unrelated arithmetic leaves. $\square$

3. APPLICATION WITH THE ARITHMETIC LABELS RETAINED

For a prime $p = 4m + 1$, let $m + 1 \leq a \leq 3m$, $R = 4a - p$, and $u = \prod q^{f_q}$ where $a = \prod q^{e_q}$ and $0 \leq f_q \leq 2e_q$. The inherited two-channel rational triples are

$$Q_E = \left(a, \frac{pa + a^2/u}{R}, \frac{pa + p^2u}{R} \right), \quad Q_M = \left(a, \frac{p(a + a^2/u)}{R}, \frac{p(a + u)}{R} \right).$$

Both satisfy the reciprocal equation. The two latter coordinates have common minimal denominator

$$D_E = R / \gcd(R, 4u + 1), \quad D_M = R / \gcd(R, u + a).$$

For E this follows by multiplying the first numerator by the unit u modulo R , obtaining $a^2(4u + 1)$; the other numerator is congruent to $4a^2(4u + 1)$. For M the same calculation has factor $u + a$. The denominator is odd, so every candidate gives the affine cover above. If g is its gate and $h = \gcd(R, g)$, the explicit group isomorphism is

$$\langle [g]_R \rangle \longrightarrow \langle [b_D] \rangle, \quad [kg]_R \longmapsto k[b_D].$$

Its inverse reads k as minus the third cusp evaluation and multiplies by g modulo R . Conversely the inverse from a residue v to k is $(v/h)(g/h)^{-1} \bmod D$.

Every integral ES solution is caught by this full interval: choose a smallest denominator a , use $p/4 < a \leq 3p/4$, and factor $(RY - pa)(RZ - pa) = p^2a^2$. Its three possible p -valuations give respectively $E, u = a^2/(RY - pa)$; $M, u = pa^2/(RY - pa)$; or the reversed exterior with $u = (RY - pa)/p^2$. Thus ES at p is equivalent to $D = 1$ for some candidate. Retain the exterior order bit, middle complementation $u \mapsto a^2/u$, all exponents, and p in the decoration. By Theorem 1.1, the disjoint union of these decorated covers has a section exactly when ES holds. There is no claim that forgetting the decoration remains injective, nor that the resulting compact curves are six-spheres.

For the restricted empty shell $p = 241, a = 64, R = 15$, each channel has return levels 3, 5, 15 with multiplicities 4, 6, 3. Keeping both exterior orders and the original middle order gives 39 states, hence 32,949 sheets in 60 compact components, with total genus 5,448. Other shells at that prime succeed. The standard-library verifier includes all these records and the full atlases for $p = 5, 37, 241, 1201$; the all-level theorems above are written proofs, not extrapolations from those finite replays.

REFERENCES

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